

Multiplying & Dividing Integers

Sign rules, computation, order of operations, and word problems

25 practice questions with step-by-step answers. Sign rules for products & quotients, multi-digit computation, order of operations with signed integers, and real-life word problems involving temperature, elevation, debt, and rates of change.

Course	Young Fermats — Pre-Algebra (grades 5–6, ages 10–12)
Class	Class 2 of 48 in the rolling YF Pre-Algebra syllabus
Meets	Small groups (max 6) · 120 minutes per week · 226 W 79th St, UWS NYC
Format	Live in-person, taught by cofounder Marcelo Ramos (Northwestern)

Free at schoolofmath.us/posts/pre-algebra-class-2-multiplying-dividing-integers

The two rules you must know cold

Every question in this class relies on **two sign rules**. They apply to BOTH multiplication and division.

SAME signs positive $\times$ positive · negative $\times$ negative	result is POSITIVE (+)
DIFFERENT signs positive $\times$ negative · negative $\times$ positive	result is NEGATIVE (-)

Shortcuts for products with 3 or more factors

Instead of applying the sign rule step by step, just count the number of **NEGATIVE** factors:

EVEN number of negatives	→ product is POSITIVE
ODD number of negatives	→ product is NEGATIVE
Any factor is 0	→ product is 0

Two rules students always forget

1. Division by 0 is undefined. Something like $(-15) \div 0$ has no answer. But $0 \div (-15) = 0$ is fine. Zero on TOP is OK; zero on the BOTTOM is not.

2. Parentheses matter with exponents. $(-3)^2 = (-3)(-3) = +9$. But -3^2 (no parentheses) $= -(3^2) = -9$. The negative sign is applied AFTER the squaring when there are no parentheses.

25 Practice Questions

Work each question on paper first. The step-by-step explanations that follow show the sign rule and the arithmetic side by side.

SIGN RULES & QUICK COMPUTATION

Q1

Fill in each blank with + or -: (a) (positive) $\times$ (negative) = ____ (b) (negative) $\times$ (negative) = ____ (c) (negative) $\div$ (positive) = ____ (d) (negative) $\div$ (negative) = ____

Answer: (a) - (b) + (c) - (d) +

Why. The two rules for signs: SAME signs give a POSITIVE result, DIFFERENT signs give a NEGATIVE result. This is true for both multiplication and division. So (a) different signs $\rightarrow$ negative. (b) same signs $\rightarrow$ positive. (c) different signs $\rightarrow$ negative. (d) same signs $\rightarrow$ positive.

Q2

Compute in your head: (a) $(-4) \times 6$ (b) $(-3) \times (-9)$ (c) $8 \times (-7)$ (d) $(-5) \times (-5)$

Answer: (a) -24 (b) 27 (c) -56 (d) 25

Why. Multiply the digits, then apply the sign rule. (a) $4 \times 6 = 24$, different signs $\rightarrow -24$. (b) $3 \times 9 = 27$, same signs $\rightarrow +27$. (c) $8 \times 7 = 56$, different signs $\rightarrow -56$. (d) $5 \times 5 = 25$, same signs $\rightarrow +25$.

Q3

Compute: (a) $(-48) \div 6$ (b) $(-72) \div (-8)$ (c) $54 \div (-9)$ (d) $0 \div (-15)$

Answer: (a) -8 (b) 9 (c) -6 (d) 0

Why. Divide the digits, then apply the sign rule. (a) $48 \div 6 = 8$, different signs $\rightarrow -8$. (b) $72 \div 8 = 9$, same signs $\rightarrow +9$. (c) $54 \div 9 = 6$, different signs $\rightarrow -6$. (d) 0 divided by any nonzero number is always 0 — no sign needed.

Q4

True or false: $(-15) \div 0 = 0$. Explain.

Answer: False. Division by 0 is UNDEFINED.

Why. You can never divide by 0. Ask yourself "what number times 0 gives -15?" — no such number exists, because anything times 0 is 0. So $-15 \div 0$ has no answer at all; we say it is undefined. Do not confuse this with $0 \div (-15) = 0$, which IS a valid statement.

Q5

Without computing, decide if each product is POSITIVE, NEGATIVE, or ZERO: (a) $(-2) \times (-3) \times (-4)$
(b) $(-1) \times (-1) \times (-1) \times (-1)$ (c) $(-7) \times 5 \times 0 \times (-2)$ (d) $(-2) \times (-3) \times 4 \times (-5)$

Answer: (a) negative (b) positive (c) zero (d) negative

Why. Count how many NEGATIVE factors there are. An EVEN count of negatives $\rightarrow$ positive product. An ODD count $\rightarrow$ negative product. If ANY factor is 0, the whole product is 0. (a) 3 negatives (odd) $\rightarrow$ negative. (b) 4 negatives (even) $\rightarrow$ positive. (c) contains 0 $\rightarrow$ zero. (d) 3 negatives (odd) $\rightarrow$ negative.

Q6

Compute: (a) $(-1)^7$ (b) $(-1)^{100}$ (c) $(-2)^3$ (d) $(-2)^4$

Answer: (a) -1 (b) 1 (c) -8 (d) 16

Why. For (-1) raised to a power: ODD exponent $\rightarrow -1$, EVEN exponent $\rightarrow +1$. So $(-1)^7 = -1$ (7 is odd) and $(-1)^{100} = 1$ (100 is even). For (-2) : $(-2)^3 = (-2)(-2)(-2) = 4 \cdot (-2) = -8$. $(-2)^4 = (-2)(-2)(-2)(-2) = 4 \cdot 4 = 16$ (four negatives make positive).

Q7

Which is bigger: $(-3)^2$ or -3^2 ? Show why.

Answer: $(-3)^2 = 9$ is bigger. $-3^2 = -9$.

Why. The parentheses change EVERYTHING. $(-3)^2$ means $(-3) \times (-3) = 9$ (positive, because same signs). But -3^2 with no parentheses means $-(3^2)$ — you square 3 first, THEN apply the negative sign. So $-3^2 = -(9) = -9$. This is one of the most common mistakes on tests — always look for the parentheses.

Q8

Fill in the missing number: (a) $(-6) \times \underline{\quad} = -42$ (b) $\underline{\quad} \div (-4) = 9$ (c) $(-8) \times \underline{\quad} = 56$

Answer: (a) 7 (b) -36 (c) -7

Why. Work backwards. (a) $42 \div 6 = 7$ (digits). Signs: negative $\times$? = negative, so ? must be POSITIVE. Answer 7. (b) $9 \times (-4) = -36$ (check: $-36 \div -4 = 9$ ✓). (c) $56 \div 8 = 7$. Signs: negative $\times$? = positive, so ? must be NEGATIVE. Answer -7.

PRODUCTS & QUOTIENTS WITH MULTI-DIGIT INTEGERS**Q9**

$(-13) \times 12 = ?$

Answer: -156

Why. $13 \times 12 = 156$ (compute normally: $13 \times 10 = 130$, $13 \times 2 = 26$, $130 + 26 = 156$). Different signs $\rightarrow$ negative. Final: -156.

Q10

$(-25) \times (-16) = ?$

Answer: 400

Why. Same signs $\rightarrow$ positive. $25 \times 16 = 400$ ($25 \times 16 = 25 \times 4 \times 4 = 100 \times 4 = 400$). Final: +400.

Q11

$(-144) \div (-12) = ?$

Answer: 12

Why. Same signs $\rightarrow$ positive. $144 \div 12 = 12$ (since $12 \times 12 = 144$). Final: +12.

Q12

$225 \div (-15) = ?$

Answer: -15

Why. Different signs $\rightarrow$ negative. $225 \div 15$: notice $15 \times 15 = 225$, so $225 \div 15 = 15$. Final: -15.

Q13

$$(-8) \times 25 \times (-4) = ?$$

Answer: 800

Why. Two negatives (even count) $\rightarrow$ positive result. Regroup for easy multiplication: $25 \times (-4) = -100$, then $(-8) \times (-100) = 800$. Or: $(-8) \times (-4) = 32$, then $32 \times 25 = 800$.

Q14Evaluate: $(-2)^5$ **Answer: -32**

Why. $(-2)^5 = (-2)(-2)(-2)(-2)(-2)$. Group in pairs: $(-2)(-2) = 4$, $(-2)(-2) = 4$. So we have $4 \times 4 \times (-2) = 16 \times (-2) = -32$. Or use the shortcut: $2^5 = 32$, and an ODD number of negatives $\rightarrow$ negative. So $(-2)^5 = -32$.

Q15

$$(-360) \div (-9) \div (-5) = ? \text{ (compute left to right)}$$

Answer: -8

Why. Division is LEFT TO RIGHT, one step at a time. Step 1: $(-360) \div (-9) = 40$ (same signs $\rightarrow$ positive; $360 \div 9 = 40$). Step 2: $40 \div (-5) = -8$ (different signs $\rightarrow$ negative; $40 \div 5 = 8$). Final: -8.

Q16Fill in the missing number: $(-234) \div \underline{\hspace{1cm}} = -18$ **Answer: 13**

Why. Signs first: negative $\div$? = negative, so ? must be POSITIVE. Digits: $234 \div ? = 18$, so ? = $234 \div 18 = 13$. Check: $(-234) \div 13 = -18$ ✓ (since $13 \times 18 = 234$).

ORDER OF OPERATIONS WITH SIGNED INTEGERS**Q17**

$$\text{Evaluate: } (-3) \times 4 + (-2) \times (-5)$$

Answer: -2

Why. PEMDAS: do the multiplications FIRST, then add. $(-3) \times 4 = -12$. $(-2) \times (-5) = +10$. Now add: $-12 + 10 = -2$. Final: -2.

Q18

Evaluate: $20 \div (-4) - (-6) \times 3$

Answer: 13

Why. PEMDAS: division and multiplication first, left to right. $20 \div (-4) = -5$ (different signs). $(-6) \times 3 = -18$ (different signs). Now the expression becomes $-5 - (-18) = -5 + 18 = 13$. Final: +13.

Q19

Evaluate: $(-2)^3 \times 5 - (-4)^2$

Answer: -56

Why. PEMDAS: exponents first, then multiplication, then subtraction. $(-2)^3 = -8$ (odd exponent $\rightarrow$ negative). $(-4)^2 = 16$ (even exponent $\rightarrow$ positive). Now: $-8 \times 5 = -40$. $-40 - 16 = -56$. Final: -56.

Q20

Evaluate: $(-12) + 4 \times (-3) \div 6$

Answer: 4

Why. Parentheses first: $(-12) + 4 = -8$. Then left to right: $(-8) \times (-3) = 24$ (same signs). Finally $24 \div 6 = 4$. Final: +4.

WORD PROBLEMS — REAL-LIFE SIGNED PRODUCTS & QUOTIENTS**Q21**

The temperature drops 3°F every hour. If it is 5°F right now, what will the temperature be 6 hours from now?

Answer: -13°F

Why. "Drops 3°F per hour" means the change is $-3^\circ\text{F}/\text{hour}$. Over 6 hours, total change = $-3 \times 6 = -18^\circ\text{F}$. New temperature = $5 + (-18) = -13^\circ\text{F}$.

Q22

A submarine is diving at 12 meters per minute. It started at sea level (0 m). Where is it 8 minutes later? (positive = above sea level, negative = below)

Answer: -96 meters (96 m below sea level)

Why. Diving at 12 m per minute means the rate is -12 m/min. Over 8 minutes, total change = $-12 \times 8 = -96$ m. Starting at 0, new depth = $0 + (-96) = -96$ m.

Q23

A stock lost \$84 in value over 7 days at a steady rate. What was the average change per day?

Answer: -\$12 per day

Why. "Lost \$84" means total change = -\$84. Steady rate over 7 days: average = $-84 \div 7 = -\$12$ per day. The negative sign tells us the stock LOST value on average.

Q24

Amara owes her brother \$6. She owes him for 4 more toys at the same price. What integer represents her total debt after buying the extra toys?

Answer: -\$30 (she owes \$30 total)

Why. Debt is represented by NEGATIVE numbers. She already owes -\$6 (that's 1 toy). Four more toys at -\$6 each = $4 \times (-6) = -\$24$. Total debt = $-6 + (-24) = -\$30$. She owes her brother \$30 total.

Q25

A hot-air balloon is descending at a rate of 8 feet per second. If it needs to descend 480 feet to land, how many seconds until it lands?

Answer: 60 seconds (1 minute)

Why. "Descending 8 feet per second" is a rate of -8 ft/sec. Total descent needed = -480 ft. Time = distance $\div$ rate = $(-480) \div (-8) = +60$ seconds. Same signs $\rightarrow$ positive time, which makes physical sense (time can't be negative). The balloon lands after 60 seconds, or 1 minute.

About SOMATH & the Young Fermats course

SOMATH (School of Math) is a math-focused school on the Upper West Side of Manhattan, cofounded by **Marcelo Ramos** (Northwestern-trained lead math teacher) and **Vivianne Wright** (Harvard-trained, also runs MBA House, one of the largest international GMAT/EA prep firms). Every class at SOMATH is small-group in-person — maximum 6 students, 120 minutes per week — held in our classroom at 226 W 79th Street, between Amsterdam and Broadway.

Young Fermats — Pre-Algebra is our 48-class rolling program for grade 5–6 students (ages 10–12). It runs from integers on the number line through systems of linear equations and probability — the direct on-ramp to Algebra 1 in 7th or 8th grade. Students can start any Monday because the syllabus is rolling: the class order is designed so that no class strictly depends on the previous week.

How to enroll

Every new family starts with a **free 60-minute evaluation** in our 226 W 79th Street classroom, plus a written diagnostic delivered within 48 hours. The diagnostic maps your child's current level against the readiness signals for Pre-Algebra and Algebra 1 — and gives you a clear next step whether you enroll or not.

Book online: schoolofmath.us/evaluation

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